

Paper Trading Bot — Daily Range Breakout Strategy

Date: 2026-08-19

Status: Approved — ready for implementation

Asset scope: ETH/USD and SOL/USD

Starting capital: \$100 per asset (paper)

1. Goal

Build a paper trading bot that trades both ETH and SOL using the Daily Range Breakout strategy on 5-minute candles. The bot runs on real Binance market data, executes simulated trades, and logs every trade with ML-ready features — laying the groundwork for a future real-money deployment backed by machine learning.

2. Architecture

2.1 Backend: `PaperTrader` class (in `dashboard.py`)

One instance per asset (`PaperTrader("ETH")`, `PaperTrader("SOL")`). Each instance:

- Receives 5m candle arrays from the existing `/api/klines` fetch pipeline
- Manages its own state: balance, open position, daily range, trade history
- Computes entry/exit signals on every candle update
- Appends every trade to `paper_trades.json`
- Exposes state to the frontend via `self.state["paper_eth"]` / `self.state["paper_sol"]`

2.2 Frontend: Chart overlay + PnL panel

- Range lines rendered on the existing Lightweight Charts candlestick chart
- Trade entry/exit markers placed on candles
- Compact PnL strip below the chart
- Toggle controls for range mode and bot on/off

2.3 Persistent log: `paper_trades.json`

Append-only JSON file. Structure designed for future ML training data ingestion.

3. Strategy Rules

3.1 Range Definition

Two modes, user-toggable via UI button:

ORB mode (default — Opening Range Breakout): - At UTC 00:00, collect the first $24 \times 5m$ candles (2 hours: 00:00–02:00 UTC) - Range high = highest high across those 24 candles - Range low = lowest low across those 24 candles - Range height = $\text{range_high} - \text{range_low}$ - If dashboard starts mid-day, backfill from today's existing 5m candles up to 24 - Range resets at next UTC 00:00

Prev Day mode: - Range high = yesterday's actual high (highest high of all 288 5m candles) - Range low = yesterday's actual low (lowest low of all 288 5m candles) - Available immediately at UTC 00:00 with no 2-hour wait

3.2 Entry Signals

- **Long entry:** 5m candle closes above range_high → buy at close price
- **Short entry:** 5m candle closes below range_low → sell at close price
- Only one position at a time per asset (no stacking)
- **Cooldown:** No new trade for 3 candles (15 minutes) after closing a position — prevents whipsaw re-entries

3.3 Position Sizing

- **All-in:** Use entire balance for each trade
- $\text{Qty} = \text{balance} / \text{entry_price}$ (fractional shares — e.g. 0.0317 ETH)

3.4 Exit Rules — Scale-out (2 legs)

Leg 1 (50% of position): - Long TP1 = $\text{entry_price} + (\text{range_height} \times 1.5)$ - Short TP1 = $\text{entry_price} - (\text{range_height} \times 1.5)$ - On hit: close 50% of position, record partial PnL, update balance

Leg 2 (remaining 50%): - Trail with $2 \times \text{ATR}(14)$ from best price since entry - Long $\text{trail_stop} = \text{highest_close_since_entry} - (2 \times \text{ATR}(14))$ - Short $\text{trail_stop} = \text{lowest_close_since_entry} + (2 \times \text{ATR}(14))$ - Updates every candle. On hit: close remainder, record final PnL

Emergency stop loss: - Hard SL at opposite side of range - Long SL = range_low - Short SL = range_high - If hit before any TP, close full position at SL price

3.5 ATR Calculation

- ATR(14) computed from 5m candles
- Uses existing Binance kline data (high, low, close)
- Calculated on the last 14 candles at the moment of entry

4. Chart Visualization

4.1 Range Lines

- Two horizontal lines using `addLineSeries()`:
- **Range High:** cyan dashed (`#22d3ee`, `lineStyle: 2`)
- **Range Low:** amber dashed (`#f59e0b`, `lineStyle: 2`)
- Semi-transparent fill zone between the lines (shaded rectangle)
- Lines extend from range start time to right edge of chart
- Old range lines fade to 30% opacity when new range resets

4.2 Trade Markers

Using candlestick series `.setMarkers()` API: - **Entry long:** green upward arrow (▲) below candle, text: "LONG @ \$3,245" - **Entry short:** red downward arrow (▼) above candle, text: "SHORT @ \$3,180" - **TP1 hit:** cyan diamond (◆) on candle, text: "TP1 +\$2.28" - **TP2 / trail close:** cyan cross (X) on candle, text: "TRAIL +\$3.03" - **Stop loss:** red cross (X) on candle, text: "SL -\$4.12"

4.3 PnL Panel

Compact strip below the candlestick chart:



balance \$98.23 | PnL \$-1.77 (-1.8%) | W 3 L 2 | 60% | trades 5 | open: LONG 0.031 ETH @

- Color-coded: green positive, red negative
- Updates in real-time as candles arrive
- ETH and SOL each have their own PnL panel in their respective Market Context cards

4.4 UI Controls

Added to the Market Context card hero section: - **Range mode toggle:** [ORB (2h)] [Prev Day] — next to existing TF buttons - **Paper bot toggle:** [Paper ON] / [Paper OFF] — pauses trading without losing state - When OFF: range lines still display, but no new trades are taken

5. Persistent Log — `paper_trades.json`

5.1 File Structure

```
{
  "meta": {
    "started": "2026-08-19T00:00:00Z",
    "version": 1,
    "strategy": "daily_range_breakout",
    "timeframe": "5m"
  },
  "eth": {
    "balance": 98.23,
    "starting_balance": 100.00,
    "trades": [ ... ],
    "stats": {
      "total_trades": 5,
      "wins": 3,
      "losses": 2,
      "pnl": -1.77,
      "pnl_pct": -1.77,
      "max_drawdown": 3.2,
      "avg_win": 3.85,
      "avg_loss": -4.62,
      "win_rate": 0.60,
      "profit_factor": 1.25,
      "sharpe_approx": 0.42
    }
  },
  "sol": {
    "balance": 102.15,
    "starting_balance": 100.00,
    "trades": [ ... ],
    "stats": { ... }
  }
}
```

5.2 Per-Trade Record Schema

```
{
  "id": "eth_1724006400_0",
  "asset": "ETH",
  "direction": "long",
  "entry_ts": 1724006400,
  "entry_price": 3245.50,
  "range_high": 3260.00,
  "range_low": 3210.00,
  "range_height": 50.00,
  "range_mode": "orb",
  "atr_at_entry": 28.5,
  "qty": 0.0308,
  "leg1_exit_ts": 1724007300,
```

```

"leg1_exit_ts": 1724007300,
"leg1_exit_price": 3320.50,
"leg1_pnl": 2.28,
"leg2_exit_ts": 1724008200,
"leg2_exit_price": 3345.00,
"leg2_pnl": 3.03,
"total_pnl": 5.31,
"pnl_pct": 5.31,
"exit_reason": "trail_stop",
"candles_held": 14,
"market_regime": "trending_up",
"volatility_regime": "expanding",
"volume_at_entry": 1250,
"rsi_at_entry": 58.2,
"price_vs_range": 0.3,
"hour_bucket": 2,
"consecutive_wins": 2,
"consecutive_losses": 0
]

```

5.3 ML Feature Definitions

Feature	Type	Description
range_height	float	Height of the daily range in USD
range_mode	string	"orb" or "prev_day"
atr_at_entry	float	ATR(14) at time of entry
volume_at_entry	float	Volume on entry candle
rsi_at_entry	float	RSI(14) at time of entry
price_vs_range	float	0.0 = at range_low, 1.0 = at range_high (where entry happened)
hour_bucket	int	0-5, which 4h window (0=00-04, 1=04-08, ...)
market_regime	string	"trending_up" / "trending_down" / "ranging" (from 1h 20/50 EMA)

<code>volatility_regime</code>	string	"expanding" / "normal" / "compressed" (ATR14 vs 50-period ATR avg)
<code>consecutive_wins</code>	int	Current win streak
<code>consecutive_losses</code>	int	Current loss streak
<code>candles_held</code>	int	How many 5m candles the position was open
<code>exit_reason</code>	string	"tp1" / "trail_stop" / "stop_loss"

6. ML Readiness

The JSON schema is designed so a future ML pipeline can:

1. **Ingest** `paper_trades.json` → pandas DataFrame
2. **Feature columns** are pre-computed per trade (no feature engineering needed at training time)
3. **Labels:** `total_pnl`, `pnl_pct`, win/loss binary, risk-reward achieved
4. **Backward compatible:** Adding new fields later won't break existing records — use `trade.get("field", default)`

Future models (not built now): - Signal filter: predict probability of profitable breakout → skip or trade - Exit optimizer: predict optimal TP multiplier based on volatility regime - Position sizing Kelly: given win rate and edge, compute optimal fraction

7. Files to Create/Modify

File	Action	Purpose
<code>dashboard.py</code>	Modify	Add <code>PaperTrader</code> class, integrate into <code>snapshot()</code> , <code>klines_api()</code>
<code>static/app.js</code>	Modify	Range lines, markers, PnL panel, toggle controls

<code>static/index.html</code>	Modify	New DOM elements for PnL strip, range toggle, paper toggle
<code>static/style.css</code>	Modify	Styles for range lines, markers, PnL panel
<code>paper_trades.json</code>	Create	Persistent trade log (auto-created on first trade)

8. Constraints

- Paper trades only — no real money, no wallet interaction
- All fills at candle close price (no slippage modeled in v1)
- One position per asset at a time
- Bot runs only while dashboard server is running (no standalone process)
- Existing candlestick chart and all other dashboard features remain unchanged
- Range lines and markers overlay on top of existing candles without replacing them